

Linux Troubleshooting Commands – Complete Guide

1. netstat -na

- netstat: Shows active network connections and ports.
- -n: Shows numeric addresses instead of resolving hostnames.
- -a: Shows all listening and established connections.

2. ss -tulnp

- ss: Replacement for netstat, faster and more detailed.
- -t: TCP ports
- -u: UDP ports
- -l: Listening ports
- -n: Numeric output
- -p: Show process using the port

3. top / htop

- Shows running processes, CPU usage, memory usage.
- htop is interactive and color-coded.

4. ps aux

- Lists all running processes with details.

5. free -h

- Shows memory usage.
- -h: Human-readable (MB/GB).

6. df -h

- Shows disk usage for all partitions.
- -h: Human-readable storage units.

7. `du -sh *`

- Shows folder/file sizes in the current directory.
- `-s`: Summary
- `-h`: Human-readable

8. `tail -f /var/log/messages`

- Shows live updates to system logs.

9. `journalctl -xe`

- Shows detailed system logs with errors.

10. `ping`

- Tests connectivity to a server.

11. `traceroute`

- Shows the path packets take to a server.

12. `ip addr`

- Shows network interfaces and IP addresses.

13. `ip route`

- Shows routing table.

14. `systemctl status`

- Checks service status.

15. `systemctl restart`

- Restarts a service.

16. `lsof -i :`

- Shows which process is using a specific port.

17. lsblk

- Lists all storage devices.

18. fdisk -l

- Shows physical disk partitions.

19. dmesg | tail

- Shows recent kernel/system messages.

20. last

- Shows login history.

This document covers commonly used Linux troubleshooting commands with simple explanations suitable for beginners and professionals.